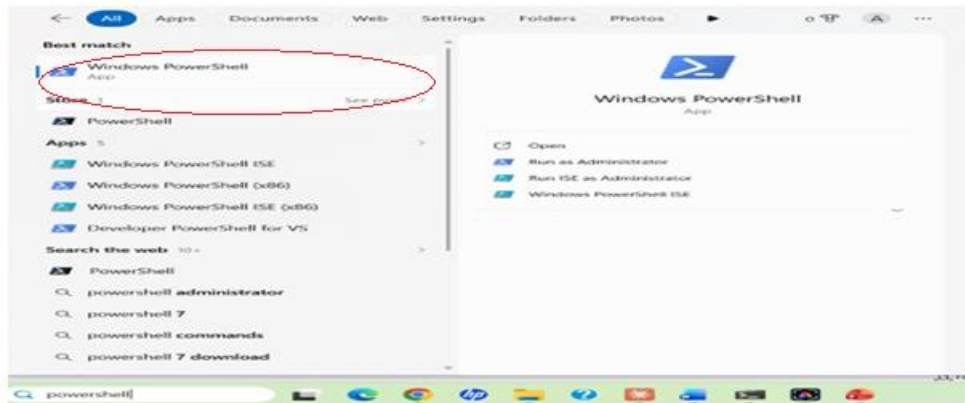
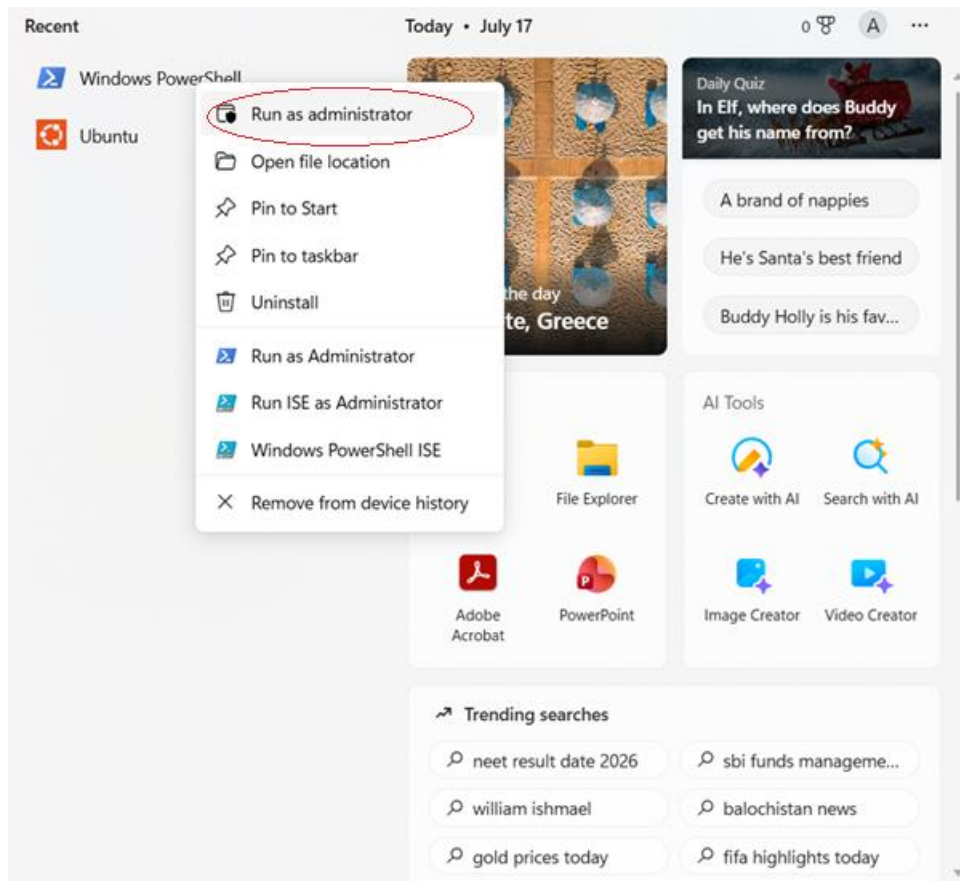


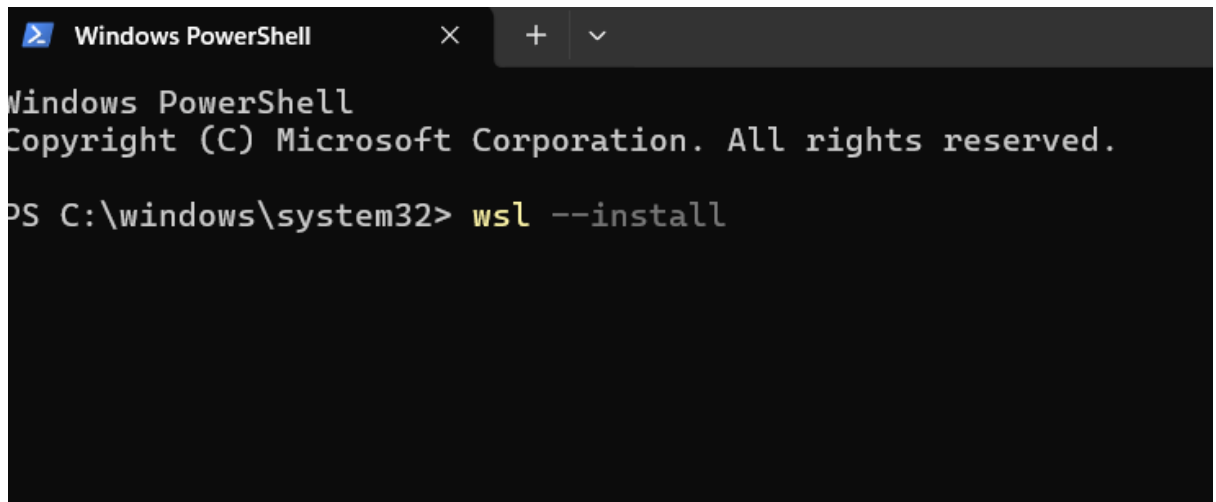
## Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal text reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', and 'PS C:\windows\system32> wsl --install'.

### Step 3: Restart Your Computer

After installation, restart your PC when prompted.

### Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: Sabiya

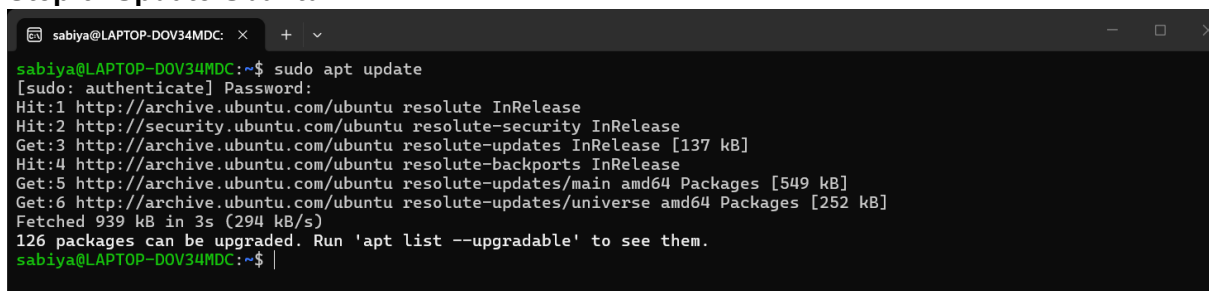
### Step 5: Create a Password

You'll then be prompted to create a password.

New password: \*\*\*\*

Retype new password:\*\*\*\*

### Step 6: Update Ubuntu

A screenshot of a terminal window titled 'sabiya@LAPTOP-DOV34MDC:'. The terminal shows the command 'sudo apt update' and its output. The output lists several Ubuntu repositories and the packages being updated. It states: '126 packages can be upgraded. Run 'apt list --upgradable' to see them.'

### Step 7: Install GCC (C Compiler)

```
sabiya@LAPTOP-DOV34MDC:~$ sudo apt install gcc
gcc is already the newest version (4:15.2.0-5ubuntu1).
gcc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 125
sabiya@LAPTOP-DOV34MDC:~$ |
```

### Step 8: Verify the Installation

```
sabiya@LAPTOP-DOV34MDC:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
sabiya@LAPTOP-DOV34MDC:~$ |
```

### Step 9: Create Your C Program

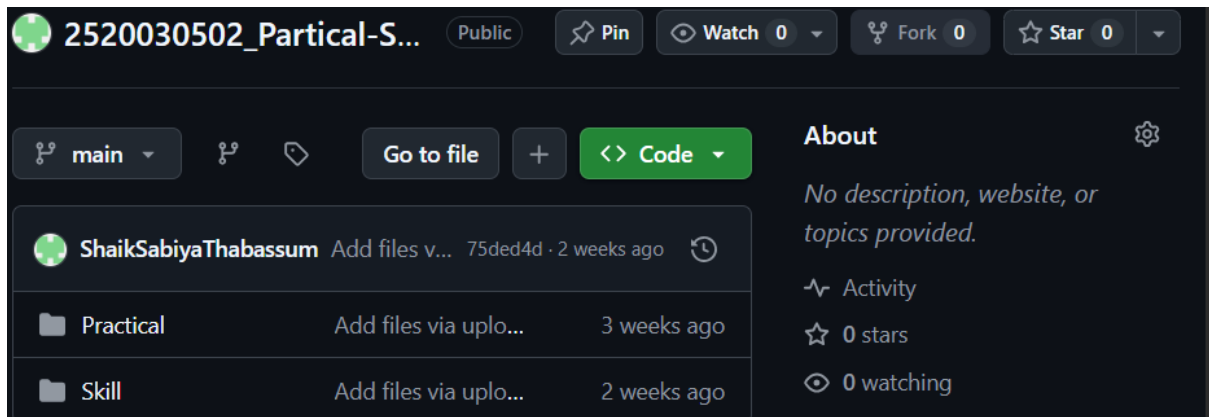
```
sabiya@LAPTOP-DOV34MDC:~$ nano process.c
sabiya@LAPTOP-DOV34MDC:~$ gcc process.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Parent Process
Parent PID: 946
Child PID: 947
Child Process
Child PID: 947
sabiya@LAPTOP-DOV34MDC:~$ |
```

Example: nano process.c

### Step 10: Compile the Program

```
sabiya@LAPTOP-DOV34MDC:~$ nano process.c
sabiya@LAPTOP-DOV34MDC:~$ gcc process.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Parent Process
Parent PID: 946
Child PID: 947
Child Process
Child PID: 947
sabiya@LAPTOP-DOV34MDC:~$ |
```

GITHUB STRUCTURE:



## EXCUTION OF FORKPROGRAMS:

### 1.Program

```
sabiya@LAPTOP-DOV34MDC:~$ nano Fork.c
sabiya@LAPTOP-DOV34MDC:~$ nano Fork.c
sabiya@LAPTOP-DOV34MDC:~$ gcc Fork.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
I am Parent
I am Child
sabiya@LAPTOP-DOV34MDC:~$ |
```

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid == 0)
    {
        printf("I am Child\n");
    }
    else
    {
        printf("I am Parent\n");
    }

    return 0;
}
```

### 2.Program

```
GNU nano 8.7.1 Frok2.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID = %d\n", getpid());
    }

    return 0;
}
```

```
sabiya@LAPTOP-DOV34MDC:~$ nano Frok2.c
sabiya@LAPTOP-DOV34MDC:~$ gcc Frok2.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Parent Process
Parent PID = 1137
Child Process
Child PID = 1138
Parent PID = 1137
sabiya@LAPTOP-DOV34MDC:~$ |
```